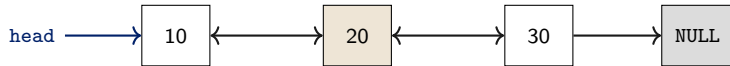
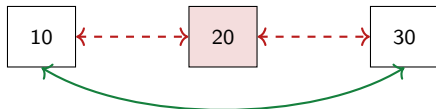


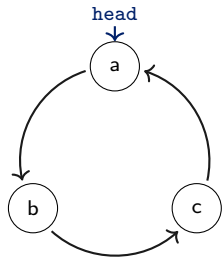
new node: 4 pointer writes



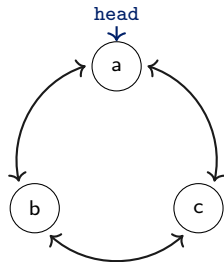
doomed: free(p)



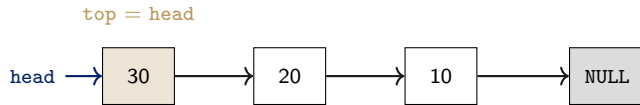
```
p->prev->next = p->next; p->next->prev = p->prev
```



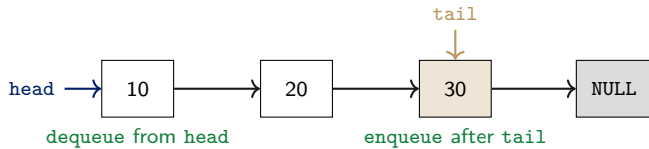
singly circular
last next = head; no NULL

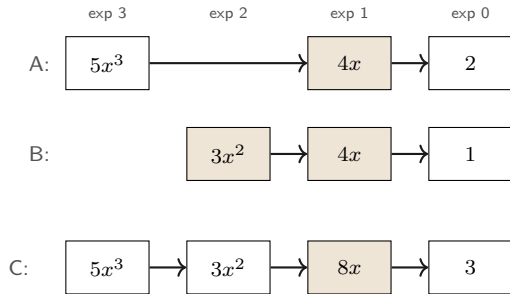


doubly circular
prev and next both close the ring



push/pop touch only head: $O(1)$ worst case





gold column: equal exponents
add coefficients ($4 + 4 = 8$), keep the exponent